MILITARY SPECIFICATION

AMMETERS, D. C. SWITCHBOARD, 8-3/4 INCH,

250-DEGREE SCALE

1. SCOPE

1.1 This specification covers the detail requirements for 250-degree scale, 8-3/4 inch direct current (d. c.) switchboard ammeters.

2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-16034 - Meters, Electrical-Indicating (Switchboard and Portable Types).

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. - The general requirements shall be in accordance with MIL-M-16034 and as specified herein.

3.2 Design, construction, and physical dimensions. - The design, construction and physical dimensions of the ammeters shall be as shown on figure 1.

3.3 Ranges. - The ammeters shall be furnished in the ranges shown in table I.

Table I - Ranges.

	Microamperes, d. c.	Milliamperes, d. c.	Amperes, d. c. 1/	Kiloamperes d. c. 1/
Full-scale ranges	200 600	1 10 100 2 20 200 5 50 500	.5 50 300 10 80 500 20 100 600 30 200 800	1 1.5 2 3

1/ Ammeters above 50 amperes and all kiloammeters are calibrated for use with 50-millivolt external shunts.

3.4 Detail requirements. - The detail requirements shall be as shown in table II.

Table II - Detail requirements.

Item	Requirement 1/
Scale length	13 inches minimum
Position influence (balance)	1 percent of scale length, maximum change
Initial accuracy	1 percent maximum error (1-1/2 percent maximum error for high-impact – shock-resistant meters)

FSC 6625

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Table II - Detail requirements, Cont'd

Item	Requirements 1/
Damping factor	3 minimum
Response time	3.5 seconds maximum
Power consumption	500 mv. maximum: milliammeters
	150 mv. maximum: self-contained ammeters
	50 mv. ± 1 percent, external-shunt type ammeter with 0.065 ± .010-ohm leads attached)
External-field influence	1 percent maximum change in 5-oersted field
Heat effect at 65°C.	4 percent maximum change at 65°C.
	0.5 percent maximum permanent change
Temperature influence	1 percent maximum change
Low-temperature exposure	2.5 percent maximum change at 0°C.
	0.5 percent maximum permanent change
Momentary overload	1 percent maximum change
Sustained overload	1 percent of scale length, maximum temporary zero shift
	0.5 percent of scale length, maximum permanent zero shift
	0.33 percent maximum permanent change in indication
Vibration	1 percent maximum change
Friction effect	2 percent maximum change
Humidity resistance	2 percent maximum change
Shock (high-impact)	5 percent maximum change

1/ Percentages are based on full-scale value, unless otherwise indicated herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. - Sampling and inspection shall be in accordance with MIL-M-16034.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery shall be in accordance with MIL-M-160334.

6. NOTES

6.1 Notes. - The notes specified in MIL-M-16034 are applicable to this specification.

Preparing activity:
Navy - SH
(Project 6625-N477Sh)

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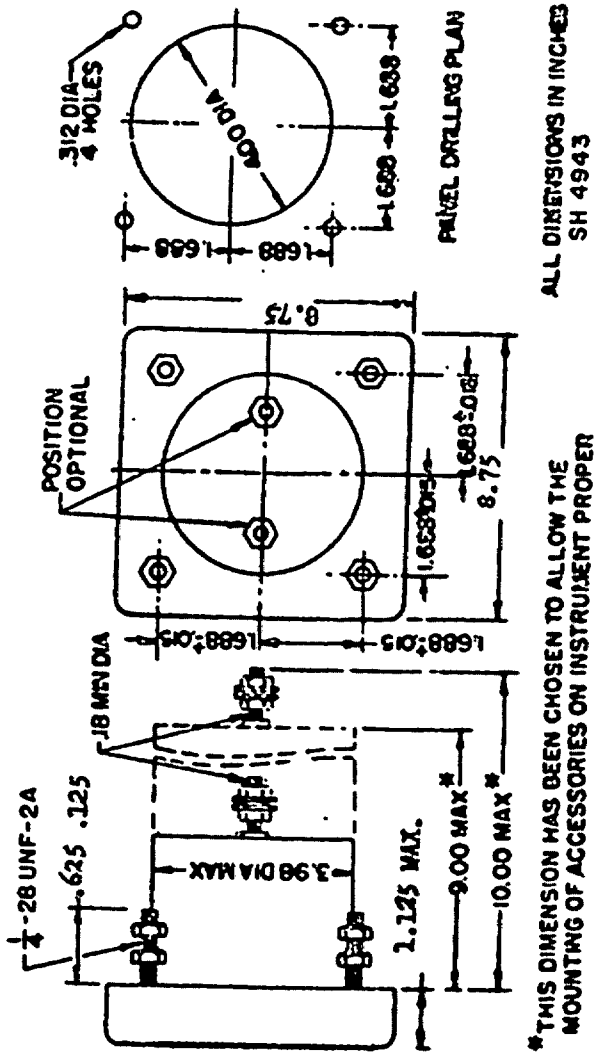


Figure 1 - Design, construction, and physical dimensions.

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